DTC P151B:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0102q1889300

DTC P151B: 00	IAC system problem
DETECTION CONDITION	• The PCM cannot control the idle speed toward the target idle speed during the KOER self test. **Diagnostic support note** • The check engine light does not illuminate. • FREEZE FRAME DATA is not available. • Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	Not applicable
POSSIBLE CAUSE	• A/C magnetic clutch malfunction • Drive-by-wire control system malfunction • Malfunction in injection amount measurement system (refer to DTC P0101:00) • Purge solenoid valve malfunction • Air cleaner element restricted • Throttle body passage blockage • Engine malfunction — Insufficient engine compression • Generator control circuit malfunction • PCM malfunction
SYSTEM WIRING DIAGRAM	Not applicable

Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
4	**INSPECT A/C MAGNETIC CLUTCH OPERATION** • Turn the A/C switch off. • Is the magnetic clutch still on?	Yes	Perform the symptom troubleshooting “NO.24 A/C IS ALWAYS ON OR A/C COMPRESSOR RUNS CONTINUOUSLY”. (See NO.24 A/C IS ALWAYS ON OR A/C COMPRESSOR RUNS CONTINUOUSLY [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
5	**INSPECT DRIVE-BY-WIRE CONTROL SYSTEM MALFUNCTION** • Perform the Drive-by-wire Control System Inspection. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 11.
		No	Go to the next step.
6	**INSPECT PURGE SOLENOID VALVE** • Perform the Purge Control System Inspection. (See ENGINE CONTROL SYSTEM OPERATION INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results, then go to Step 11. (See PURGE SOLENOID VALVE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
7	**INSPECT AIR CLEANER ELEMENT** • Remove the air cleaner element with the engine is running. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Does the engine speed increase?	Yes	Clean or replace the air cleaner element, then go to Step 11. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
8	**INSPECT THROTTLE BODY PASSAGE** • Visually inspect the throttle body passage. • Is the throttle body passage dirty and/or restricted?	Yes	Clean or replace the throttle body passage, then go to Step 11. (See INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	Go to the next step.
9	**INSPECT ENGINE COMPRESSION** • Inspect the engine compression. (See COMPRESSION INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are compression pressures within specification?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning part according to the inspection results, then go to Step 11.
10	**INSPECT GENERATOR CONTROL CIRCUIT FOR SHORT TO POWER SUPPLY** • Apply the electrical load (such as blower fan, rear window defogger, or headlight). • Does the engine speed increase?	Yes	Go to the next step.
		No	Repair or replace the malfunctioning part according to the inspection results, then go to the next step.
11	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Perform the KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) Go to the next step.
		No	Go to the next step.
12	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].)
		No	DTC troubleshooting completed.